Joseph Bramah (1749 - 1814) was source person for creating hydraulic Technology in 1795. He controlled the machine movements by using the highly pressurized liquid.

8.1 Introduction

The volume of a liquid cannot be changed by applying pressure. This property of liquid forms the base of study of hydraulics. In modern metal working plants, hydraulic pressure is being used to operate practically every type of machine tool.

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8.2 Hydraulic pumps

Hydraulic pump is a device, which transforms mechanical energy into hydraulic energy. It serves as the heart of the hydraulic circuit. It provides the force used to move the load in the circuit but the pressure of the circuit is no developed by the pump. The pressure built up in a circuit is because of the quantity of the oil or the setting of pressure relief valve.

Principle of a hydraulic pump

Every hydraulic pump has an inlet and outlet. Partial vacuum is created at the inlet port because of the rotation of the impeller. Hydraulic liquid is forced to flow from the reservoir because of this vacuum through a pipeline and is pumped out through the outlet.

Size of the pump

The size of the pump is specified by the quantity of the oil to be pumped, suction head (the depth from which oil is to be sucked) and the delivery head (the light up to which oil is to be pumped).

Types of hydraulic pump

Hydraulic pumps are classified according to three different features.

1. According to the displacement of hydraulic liquid.
   a. Positive displacement pump
   b. Non-positive displacement pump
2. According to the volume of hydraulic liquid
   a. Constant delivery pump
   b. Variable delivery pump
3. According to the nature of movement of the moving part
   a. Rotary pump
   b. Reciprocating pump

8.3 Features of positive displacement pump

1. It can pump against high pressure
2. It can handle small amount of oil at high pressure
3. When the pump is working against high pressure, the volume of the outlet will not have any slip. So the outlet should not be closed while pumping as it will cause damage to the pump. A pressure relief valve is connected to the circuit next to this pump.
4. This pump has got self priming (The pump can draw liquid from the reservoir)

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